

Business analytics, applied modelling and prediction
ST2187

Coursework

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Executive summary

This report gives an overview of the global market while also delving into the Southeast Asia region, providing insights regarding product, customer and shipping aspects of the market.

The following important information about the global market was uncovered:

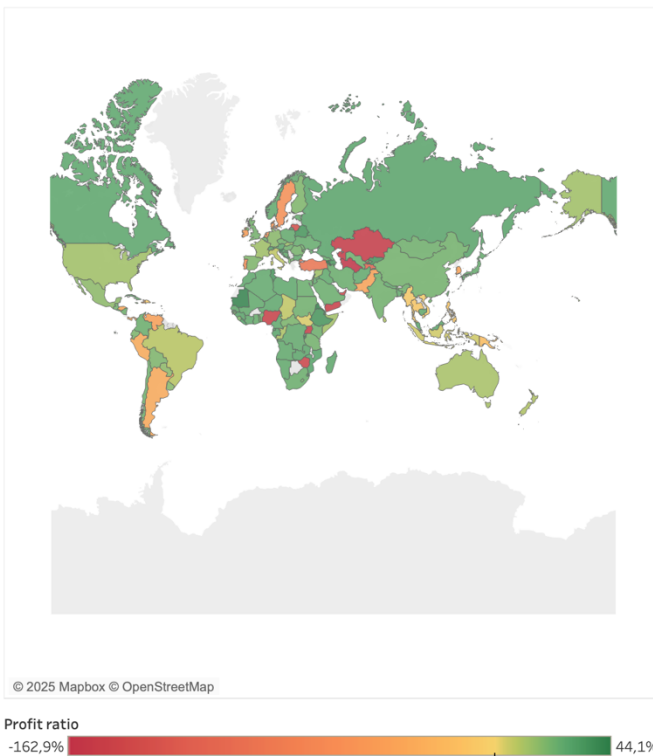
- Total profit reached 1,466,437.
- Profit ratio of 11.6%.
- Average sales per customer are 15,886.
- Profit per order averaged 59.
- The average shipping cost is 26 per order.
- The total quantity of products sold is 178,089.
- The average discount offered is 14.3%.

Overview of the global market

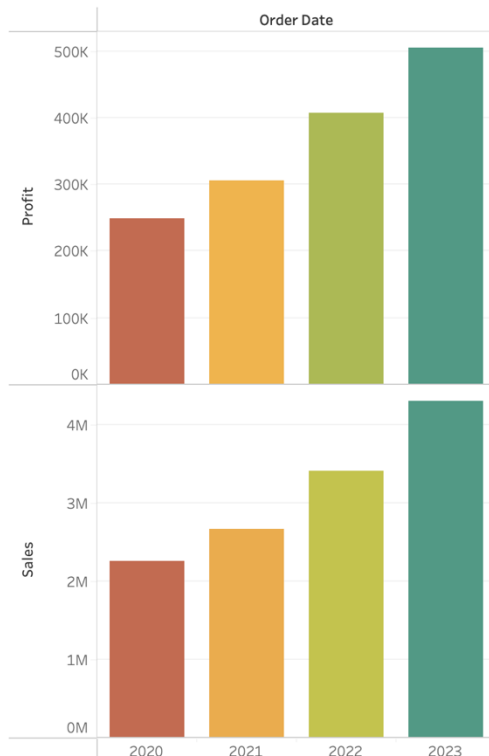
Executive summary

Sales	Sales per Customer	Profit	Profit ratio	Profit per order	Avg. Shipping Cost	Quantity	Avg. Discount
12,629,244	15,886	1,466,437	11,6%	59	26	178,089	14,3%

Profit ratio global



Profit and sales dynamic over the years



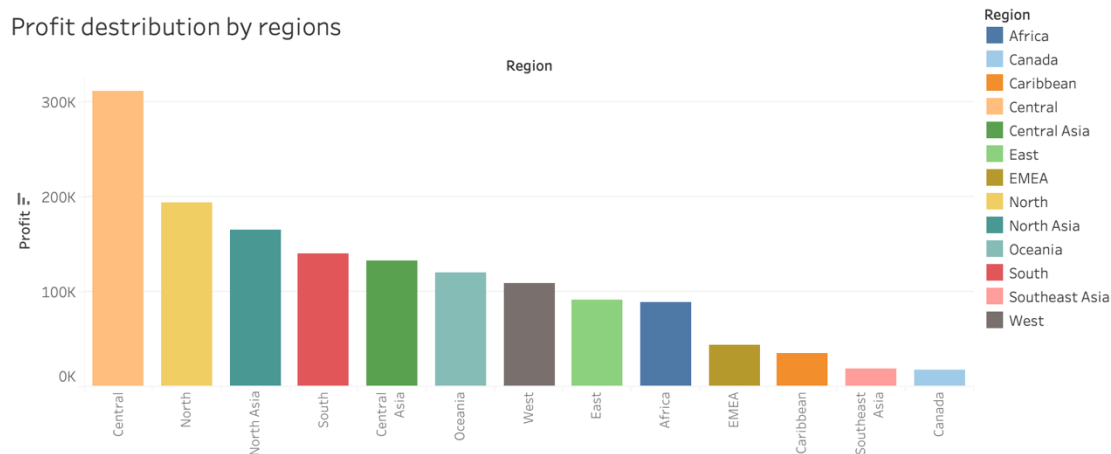
These visualizations are aimed at analyzing the global market. The first table provides an extensive summary of the core characteristics important for understanding the market's state. This information is of great analytical value for management personnel and can be used to develop further sales strategies.

The first graph gives information on profit ratios by country, highlighting countries with greater profit ratios in green and those with negative profit ratios in red. This map indicates that the overall market picture is quite good, as most territories remain in the positive profit ratio zone. Armenia has the highest profit ratio at 44.1%, along with a few countries from Africa and Europe. Areas needing attention include Turkmenistan, with a -162.9% profit ratio, and Kazakhstan, with -154.2%, both located in EMEA, and several countries in Africa.

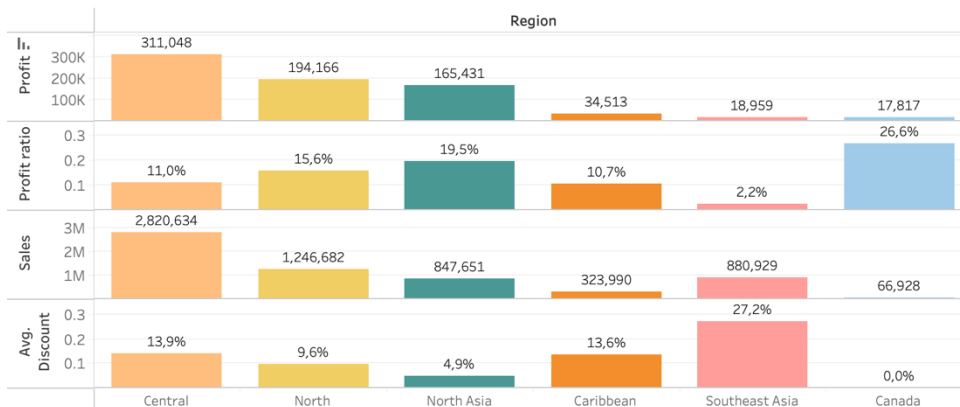
Analyzing the bar charts to the left, we observe an increase in sales and profits year by year, indicating an upward trend in the global market.

Explore top 3 most and least profitable regions

Profit distribution by regions



Top 3 most and least profitable regions



This section explores the profit distribution between regions. And the following insights were uncovered. The top 3 regions by profitability are Central – 311,048, North – 194,166 and North Asia – 165,431, while Canada – 17,817, Southeast Asia – 18,959 and the Caribbean – 34,513 are the least profitable regions.

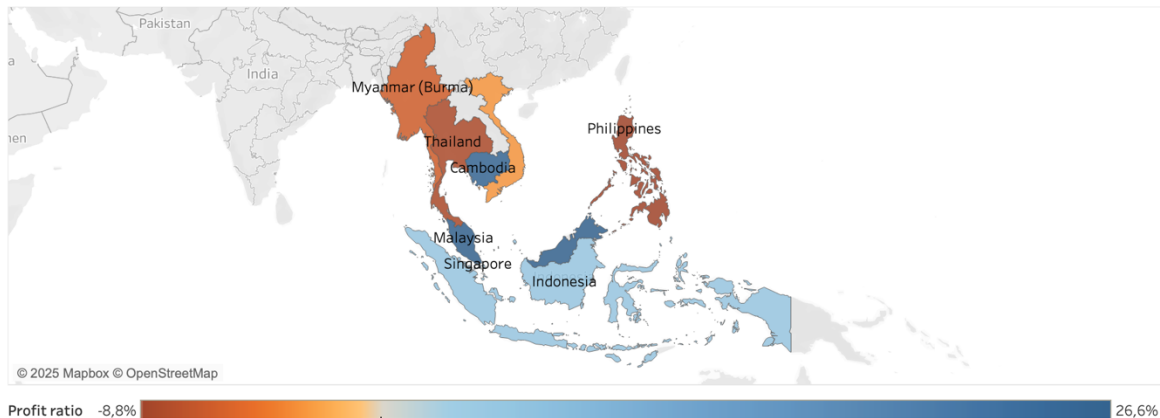
The deeper analyzes uncovers that despite North Asia being less profitable with the fewer number of sales than the Central and North regions, it excels in profit ratio among the top three regions by profitability.

While Canada surprises us with the highest profit rate – 26.6% among all 6 regions, despite the smallest number of sales – 66,928, more over the average discount bar shows that Canada is discounted at average around 0%. The above information may indicate the niche type of market in Canada.

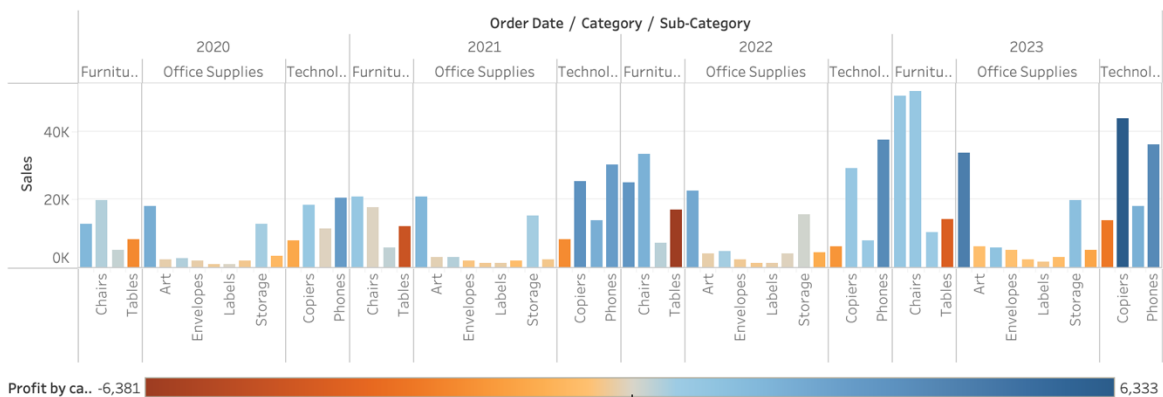
The most problematic region among the top 3 least profitable regions is Southeast Asia. Despite the fact that its number of sales can be compared with the 3rd most profitable region, the profitability of Southeast Asia is low. Moreover, it has a profit ratio near 2.2%, which is the lowest among 6 regions. Presumably, such a result may be associated with an unreasonably high average discount - 27.2%. However, in my opinion, this region requires close study, which we will delve into in the following dashboards.

Product research of the Southeast Asia market

Profit ratio of Southeast Asia



Distribution of sales by category



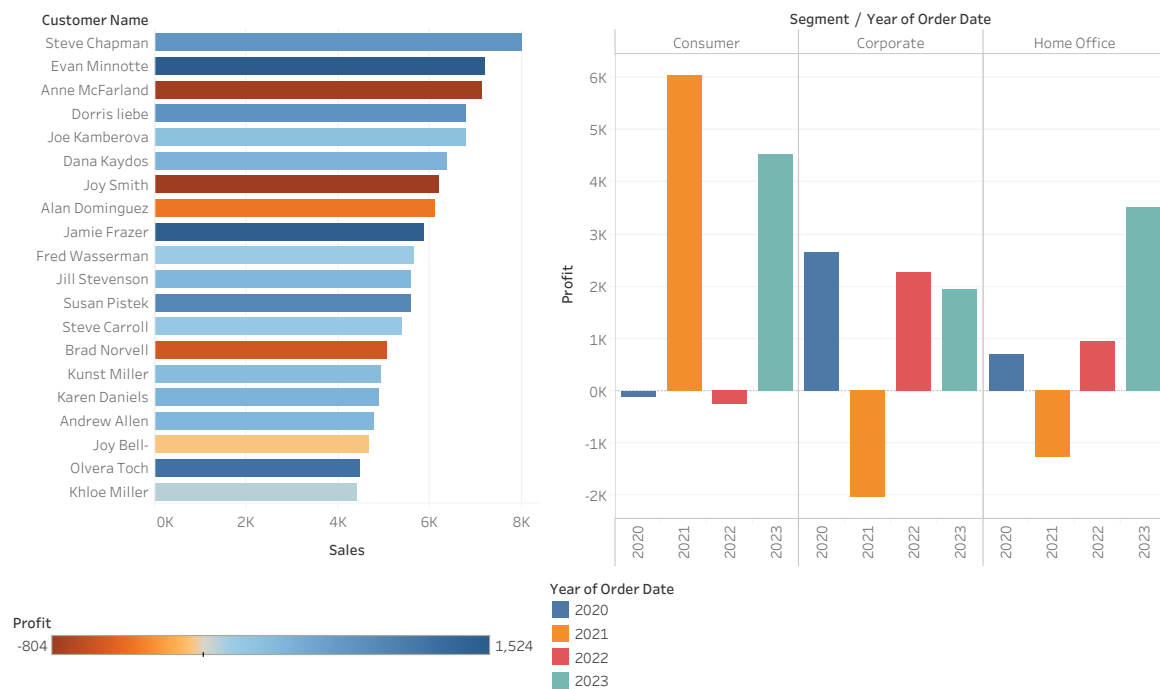
In this section you can find deeper analyses of the Southeast Asia market. The following heatmap highlights several problematic regions, such as the Philippines, Thailand, Myanmar and Vietnam, for which the profit ratio is in the negative range, which is important to pay attention to. While the profit ratio for Malaysia is 26.6% and Cambodia is 25.6%, which is a good indicator and is a potential indicator of a promising market, as they exceed the global sector average of 11.6%.

The bar chart above indicates sales distribution by category in Southeast Asia. In this visualization, we can track the recurring problems with the furniture sector, and more specifically with the subcategory tables. Despite the fact that tables are sold year after year, they bring negative profits, and by 2023 they are increasingly losing popularity. This trend may be related to the cultural characteristics of this region.

As for technologies, it shows the stable growth in sales and profit throughout years, where the two categories dominate the market, they are copiers and phones.

Customer research of the Southeast Asia market

Top 20 client with the biggest number of sales Distribution of profit by segment

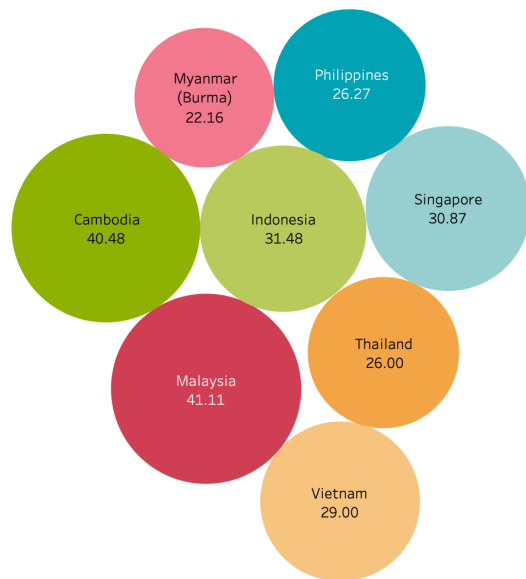


Considering customer segment research of the Southeast Asia market. The first visualization shows the top 20 customers by sales. Most of the customers who do not bring profit to the company belong to the consumer segment. While the top two customers are from the corporate and home office segment.

Moving on to the analysis of the profitability of the Southeast Asia market segments. We can see strong fluctuations in the profitability of the consumer segment, while in the corporate and home office segment a strong decline is noticeable only in 2021, and in other years the profitability remains positive. In general, it can be seen that the profitability of all segments in 2023 came to a good positive number, which can confirm the correctness of the chosen strategy.

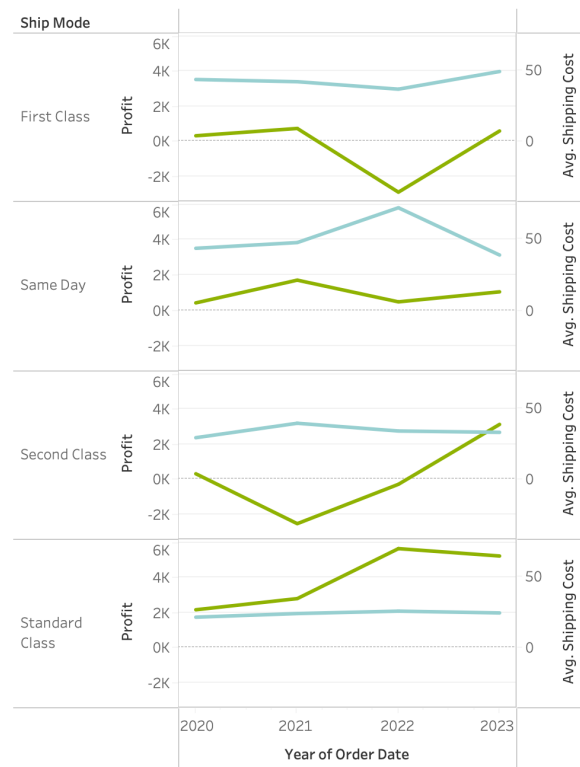
Shipping insides of the Southeast Asia

Country by average shipping cost



Measure Names
■ Avg. Shipping C... ■ Profit

Shipping cost and profit per class in Southeast Asia



In this section we will dive into product shipping. The first visualization shows countries with average shipping costs. You can see a significant difference between the average cost of shipping goods within one region. Malaysia – 41.11 and Cambodia – 40.48 are the most expensive in terms of shipping, while Myanmar – 22.16 is one of the cheapest.

The line graphs to the right show the distribution of shipping costs and profitability by shipping class. It is noticeable that in standard class profits have increased from 2020 to 2023 while the average cost remained constant. Moreover, of all the shipping classes, it is the most profitable one. The most expensive shipping classes are first and same-day class, but their high cost is not justified by their profitability, as they do not show outstanding results. Moreover, from 2021 to 2023, the yield on the first class falls to negative values, reaching a minimum of -2.888, a similar situation to the second class, which receives negative profits during 2020–2022.

Recommendations

Global suggestions:

- Focus on the countries with the higher profit ratio as it serves as the indicator of the financial health of the area.
- Address problematic areas so that the strategy in countries like Turkmenistan and Kazakhstan will be reconsidered for better efficiency..

Southeast Asia suggestions:

- Address problematic countries as the Philippines, Thailand, Myanmar and Vietnam owing to the fact that factors like profit ratio give a warning of the wrong management decisions.
- Consider reevaluating tables necessity in this region, try to discount it or address the possible cultural meaning behind this and launch a more appropriate alternative to this market.
- Focused on the more profitable categories like technologies, try to leverage the popularity of these items.
- Consider negotiating better deals with the shipping companies regarding costs for Malaysia and Cambodia as it can boost their market growth and bring more profit.
- Concentrate on the home office and corporate segments as it provides more stable profit.

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